

closed hashing:

- linear probing
- Quadratic probing

Double hashing:

A → 3, 2, 9, 6, 11, 13, 7, 12

M = 10

0	
1	9
2	
3	
4	
5	11
6	
7	7 12
8	
9	3 13

key	locate (hk) = $2k + 3$
3	$9 = 1 \cdot 10 = 9$
2	$7 = 1 \cdot 10 = 7$
9	$21 = 2 \cdot 10 = 1$
6	$15 = 1 \cdot 10 = 5$
11	$25 = 2 \cdot 10 = 5$
13	$29 = 2 \cdot 10 = 9$
7	$17 = 1 \cdot 10 = 7$
12	$24 = 2 \cdot 10 = 4$

linear probing

A → 3, 2, 9, 6, 11, 13, 7, 12 M = 10

0	13
1	9
2	12
3	
4	
5	6
6	11
7	2
8	7
9	3

key	locate	probing (u+i)
3	$9 = 1 \cdot 10 = 9$	
2	$7 = 1 \cdot 10 = 7$	
9	$21 = 2 \cdot 10 = 1$	
6	$15 = 1 \cdot 10 = 5$	
11	$25 = 2 \cdot 10 = 5 \Rightarrow 6$	$5 \Rightarrow 6$
13	$29 = 2 \cdot 10 = 9 \Rightarrow 0$	$9 \Rightarrow 0$
7	$17 = 1 \cdot 10 = 7 \Rightarrow 8$	$7 \Rightarrow 8$
12	$24 = 2 \cdot 10 = 4 \Rightarrow 2$	$4 \Rightarrow 2$

A → 8, 2, 9, 6, 11, 13, 7, 12

		key	locate (u)	probing (u+i ²)
0	13			
1	9	8	9	1
2	12	2	1	1
3		9	1	1
4		6	5	1
5	6	11	5	2
6	11	13	9	2
7	2	7	7	2
8	7	12	7	6
9	3			

$v = 4$

$(5 + 4 \times 4) \cdot 10$

$u =$
 $v =$
 $i =$

$= (u + i) \cdot 10$

~~$= (9 + 0) \cdot 10 = 9$~~

$= (5 + 0) \cdot 10 = 5$

$= (5 + 1) \cdot 10 = 6$

$= (9 + 0) \cdot 10 = 9$

$= (9 + 1) \cdot 10 = 0$

$= (7 + 0) \cdot 10 = 7$

$= (7 + 1) \cdot 10 = 8$

$= (7 + 2^2) \cdot 10 = 1$

$(7 + 9) \cdot 10 = 8$

$(7 + 16) \cdot 10 = 3$

23% 10 27

2/25 20 100%